



Homework 1

Linear Regression · Quarto Blogging

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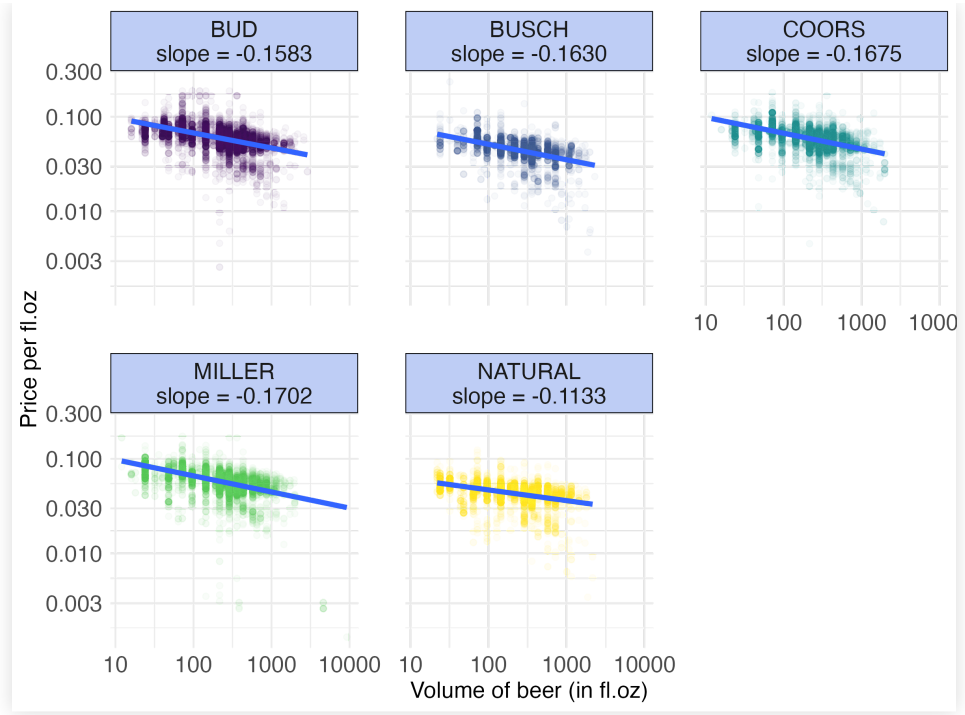
Directions

- Submit your **Quarto document** for **Part 1** to Brightspace using this filename format:
 - `danl-320-hw1-LASTNAME-FIRSTNAME.qmd`
(e.g., `danl-320-hw1-choe-byeonghak.qmd`)
- **Due:** February 11, 2026, 3:15 PM (ET)
- Questions? Email Byeong-Hak bchoe@geneseo.edu.

Required R Packages

```
library(tidyverse)
library(stargazer)
library(broom)
```

Part 1. Linear Regression



Consider the `beer_markets` data:

```
library(tidyverse)

beer_markets <- read_csv(
  "https://bcdanl.github.io/data/beer_markets_all_cleaned.csv"
)
```

Show

10

 entries

Search:

	household	X_purchase_desc	quantity	brand	dollar_spent	beer_floz	price_floz
1	2000946	BUD LT BR CN 12P	1	BUD_LIGHT	8.14	144	0.05652777777777778
2	2003036	BUD LT BR CN 24P	1	BUD_LIGHT	17.48	288	0.06069444444444445
3	2003036	BUD LT BR CN 24P	2	BUD_LIGHT	33.92	576	0.05888888888888889
4	2003036	BUD LT BR CN 30P	2	BUD_LIGHT	34.74	720	0.04825
5	2003036	BUD LT BR CN 36P	2	BUD_LIGHT	40.48	864	0.04685185185185184
6	2003036	BUD LT BR CN 36P	2	BUD_LIGHT	42.96	864	0.04972222222222222
7	2003036	BUD LT BR CN 36P	2	BUD_LIGHT	40.96	864	0.0474074074074074
8	2001521	BUD LT BR CN 6P	5	BUD_LIGHT	30.6	480	0.06375

	household	X_purchase_desc	quantity	brand	dollar_spent	beer_floz	price_floz
9	2001521	BUD LT BR CN 6P	1	BUD_LIGHT	9.99	96	0.1040625
10	2001521	BUD LT BR CN 6P	5	BUD_LIGHT	30.7	480	0.06395833333333333

Showing 1 to 10 of 73,115 entries

Previous

1

2

3

4

5

...

7,312

Next

Variable Description

Variable Name	Description
household	Unique identifier for household
X_purchase_desc	Description of beer purchase
quantity	Number of beer packages purchased
brand	Brand of beer purchased
dollar_spent	Total amount spent on the purchase
beer_floz	Total volume of beer purchased (in fluid ounces)
price_floz	Price per fluid ounce of beer
container	Type of beer container (e.g., CAN, BOTTLE)
promo	Indicates if the purchase was part of a promotion (TRUE/FALSE)
region	Geographic region of purchase
marital	Marital status of household head
income	Income level of the household
age	Age group of household head
employment	Employment status of household head
degree	Education level of household head
occupation	Occupation category of household head
ethnic	Ethnicity of household head
microwave	Indicates if the household owns a microwave (TRUE/FALSE)
dishwasher	Indicates if the household owns a dishwasher (TRUE/FALSE)

Variable Name	Description
<code>tv_cable</code>	Type of television subscription (e.g., basic, premium)
<code>single_family_home</code>	Indicates if the household is a single-family home (<code>TRUE/FALSE</code>)
<code>n_people</code>	Number of people in the household

Question 1 — Filter the data (containers)

Create a data frame that keeps **only** observations where `container` is:

- `"CAN"` or
- `"NON_REFILLABLE_BOTTLE"`

Requirements:

- Name your filtered data frame `beer_sub`.
- Report the number of observations in `beer_sub`.
- Report a frequency table of `container` in `beer_sub`.

Show answer

With `%in%` operator

```
beer_sub <- beer_markets |>
  filter(container %in% c("CAN", "NON_REFILLABLE_BOTTLE"))
```

With OR (`|`) operator

```
beer_sub <- beer_markets |>
  filter(container == "CAN" | container == "NON_REFILLABLE_BOTTLE")
```

```
nrow(beer_sub)
```

```
[1] 72110
```

```
beer_sub |>
  count(container)
```

```
# A tibble: 2 × 2
  container      n
  <chr>      <int>
1 CAN        53015
2 NON_REFILLABLE_BOTTLE 19095
```

Question 2 — Train/test split

Split `beer_sub` into two data frames:

- Training set (`beer_train`): about **67%** of observations
- Test set (`beer_test`): the remaining observations

Requirements:

- Use a random seed for reproducibility (you choose the seed).
- Report `nrow(beer_train)` and `nrow(beer_test)`.
- Briefly explain *why* we keep a test set.

Show answer

With `runif(n())`

```
# 1) reproducible split
set.seed(1) # you can choose any seed

# 2) sample row indices
beer_sub <- beer_sub |>
  mutate(rnd = runif(n())) # n() number of random numbers from Unif(0,1) are drawn

# Setting reference levels
beer_sub <- beer_sub |>
  mutate(market = factor(market),
         market = fct_relevel(market, "BUFFALO-ROCHESTER"),
         brand = factor(brand),
         brand = fct_relevel(brand, "BUD_LIGHT")
  )

# 2-1) sample row indices for training (about 67%)
beer_train <- beer_sub |>
  filter(rnd >= 0.33)

# 2-1) sample row indices for test (about 33%)
beer_test <- beer_sub |>
  filter(rnd < 0.33)

# 3) report sizes
nrow(beer_train)
```

```
[1] 48322
```

```
nrow(beer_test)
```

```
[1] 23788
```

With `sample.int(n)`

```
## eval: false

# 1) reproducible split
set.seed(1) # you can choose any seed

# 2) sample row indices for training (about 67%)
n <- nrow(beer_sub)
train_idx <- sample.int(n = n, size = round(0.67 * n), replace = FALSE)

beer_train <- beer_sub[train_idx, ]
beer_test <- beer_sub[-train_idx, ]
```

```
# 3) report sizes
nrow(beer_train)
```

```
[1] 48314
```

```
nrow(beer_test)
```

```
[1] 23796
```

Questions 3–8 (Regression models)

You will estimate three linear regression models for:

- Outcome: $\log(\text{price_floz})$

Reference levels (required):

- Set "BUD_LIGHT" as the reference level for `brand`.
- Set "BUFFALO-ROCHESTER" as the reference level for `market`.
 - There are $N + 1$ distinct categories in `market`:

Model 1

$$\log(\text{price_floz}) = \beta_0 + \sum_{i=1}^N \beta_i \text{market}_i + \sum_{j=N+1}^{N+4} \beta_j \text{brand}_j + \beta_{N+5} \text{container_CAN} + \beta_{N+6} \log(\text{beer_floz}) + \epsilon$$

Model 2 (brand-specific volume sensitivity)

$$\begin{aligned} \log(\text{price_floz}) = & \beta_0 + \sum_{i=1}^N \beta_i \text{market}_i + \sum_{j=N+1}^{N+4} \beta_j \text{brand}_j + \beta_{N+5} \text{container_CAN} \\ & + \beta_{N+6} \log(\text{beer_floz}) \\ & + \sum_{j=N+1}^{N+4} \beta_{j \times \text{beer_floz}} \text{brand}_j \times \log(\text{beer_floz}) \\ & + \epsilon \end{aligned}$$

Model 3 (promo + brand-specific promo effects)

$$\begin{aligned}
\log(\text{price_floz}) = & \beta_0 + \sum_{i=1}^N \beta_i \text{market}_i + \sum_{j=N+1}^{N+4} \beta_j \text{brand}_j + \beta_{N+5} \text{container_CAN} \\
& + \beta_{N+6} \log(\text{beer_floz}) \\
& + \beta_{N+7} \text{promo} \times \log(\text{beer_floz}) \\
& + \sum_{j=N+1}^{N+4} \beta_{j \times \text{beer_floz}} \text{brand}_j \times \log(\text{beer_floz}) \\
& + \sum_{j=N+1}^{N+4} \beta_{j \times \text{promo}} \text{brand}_j \times \text{promo} \\
& + \sum_{j=N+1}^{N+4} \beta_{j \times \text{promo} \times \text{beer_floz}} \text{brand}_j \times \text{promo} \times \log(\text{beer_floz}) \\
& + \epsilon
\end{aligned}$$

How to Interpret β Coefficients in Linear Regression

Assume d is a **dummy variable** (0/1). Interpretations are **ceteris paribus** (holding other variables constant).

1) Level–Level Model

$$y = \beta_1 x_1 + \beta_2 d$$

Interpretation

- β_1 : a **one-unit increase** in x_1 changes y by β_1 **units**.
 - “If x_1 increases by 1, expected y changes by β_1 .”
- β_2 (dummy): changing d from 0 to 1 changes y by β_2 **units**.
 - “The $d = 1$ group has y that is β_2 higher/lower than the $d = 0$ group.”

2) Log–Level Model (Semi-log)

$$\log(y) = \beta_1 x_1 + \beta_2 d$$

Interpretation (for x_1)

- β_1 : a one-unit increase in x_1 changes y by approximately $100\beta_1\%$.
 - More accurate:
 - $\% \Delta y = 100(e^{\beta_1} - 1)$

Interpretation (for dummy d)

- β_2 : switching from $d = 0$ to $d = 1$ changes y by:
- $\% \Delta y = 100(e^{\beta_2} - 1)$
 - “The $d = 1$ group has y that is $100(e^{\beta_2} - 1)\%$ higher/lower than the $d = 0$ group.”

3) Log–Log Model (Elasticity Model)

$$\log(y) = \beta_1 \log(x_1) + \beta_2 d$$

Interpretation (for $\log(x_1)$)

- β_1 is an **elasticity**:
 - a **1% increase** in x_1 is associated with a $\beta_1\%$ change in y .
 - “If x_1 rises by 1%, expected y changes by $\beta_1\%$.”

Interpretation (for dummy d)

- β_2 : switching from $d = 0$ to $d = 1$ changes y by:

$$\bullet \ \% \Delta y = 100(e^{\beta_2} - 1)$$

Quick Summary

- **Level–Level:** β = units of y per 1 unit of x
- **Log–Level:** $\beta \approx$ % change in y per 1 unit of x
- **Log–Log:** β = % change in y per 1% change in x (elasticity)
- **Dummy in log(y):** percent difference is $100(e^{\beta} - 1)\%$

Question 3 — Model intuition (conceptual)

For **each** model, explain in plain English:

- What is the model trying to capture?
- What kinds of differences across markets and brands are allowed?
- How does the model treat the relationship between volume (`beer_floz`) and price (`price_floz`)?
- What *new* flexibility is added when moving from Model 1 $\rightarrow$ 2 $\rightarrow$ 3?

Show answer

Pricing Models: How Volume and Promotions Affect Price per fl oz

All three models share the same outcome variable — $\log(\text{price_per_floz})$ — and the same core inputs: market fixed effects, brand fixed effects, a container type indicator, and package size $\log(\text{beer_floz})$. Because both price and volume are log-transformed, the coefficient on $\log(\text{beer_floz})$ is interpretable as an **elasticity**: a 1% increase in package size is associated with an $\beta_{N+6}\%$ change in price per fl oz. The models differ in how much that elasticity is allowed to vary.

Model 1: One common volume elasticity

$$\log(\text{price_per_floz}) = \beta_0 + \sum_m \beta_m \text{market}_m + \sum_b \gamma_b \text{brand}_b + \delta \text{container}_{\text{CAN}} + \eta \log(\text{beer_floz}) + \epsilon$$

Model 1 is the baseline. Markets, brands, and container type are each allowed to shift the **level** of price per fl oz up or down, but every brand shares the **same single elasticity** η with respect to package size. If $\eta < 0$, larger packs are cheaper per fl oz on average — a standard quantity-discount pattern — but that pattern is assumed identical across all brands. This is the most restrictive specification: it treats beer pricing as essentially homogeneous in how per-unit price scales with size, which can mask real differences in pricing power between premium and value brands.

Model 2: Brand-specific volume elasticity

$$\log(\text{price_per_floz}) = \beta_0 + \sum_m \beta_m \text{market}_m + \sum_b \gamma_b \text{brand}_b + \delta \text{container}_{\text{CAN}} + \eta \log(\text{beer_floz}) + \sum_b \theta_b (\text{brand}_b \times \log(\text{beer_floz})) + \epsilon$$

Model 2 adds brand-by-volume interactions, so each brand now has its own elasticity. For the reference brand the elasticity is η ; for any other brand b it is $\eta + \theta_b$. This captures the intuition that a premium brand may hold its per-unit price nearly constant as pack size grows, while a value brand might offer steeper quantity discounts. The remaining restriction is that promotional status does not affect these elasticities — on-promotion and off-promotion observations are pooled within each brand.

Model 3: Brand $\times$ promotion-specific volume elasticity

$$\begin{aligned} \log(\text{price_per_floz}) = & \beta_0 + \sum_m \beta_m \text{market}_m + \sum_b \gamma_b \text{brand}_b + \delta \text{container}_{\text{CAN}} + \eta \log(\text{beer_floz}) \\ & + \kappa (\text{promo} \times \log(\text{beer_floz})) + \sum_b \theta_b (\text{brand}_b \times \log(\text{beer_floz})) \\ & + \sum_b \pi_b (\text{brand}_b \times \text{promo}) + \sum_b \lambda_b (\text{brand}_b \times \text{promo} \times \log(\text{beer_floz})) + \epsilon \end{aligned}$$

Model 3 is the most flexible. It allows the volume elasticity to differ not only by brand but also by **promotional status**, and it lets that promotional shift vary by brand. Concretely:

	Reference brand	Brand b
No promo	η	$\eta + \theta_b$
On promo	$\eta + \kappa$	$\eta + \theta_b + \kappa + \lambda_b$

This matters because promotions often change the *composition* of pack sizes sold. A deal-seeking segment may trade up to larger packs when a value brand is on promotion, steepening its quantity-discount curve, while a premium brand’s elasticity may be largely unaffected by promo status. Model 3 can detect and quantify those differences.

Summary

Model 1 provides a clean, parsimonious benchmark with one shared volume response. Model 2 adds nuance by letting each brand have its own packaging sensitivity. Model 3 goes furthest by allowing that sensitivity to shift under promotions, and by letting the size of that shift vary by brand, making it the richest specification for understanding pricing and promotional dynamics.

Question 4 — Estimate, predict, and evaluate

Using `beer_train`:

- 1. Fit **Model 1**, **Model 2**, and **Model 3** using `lm()`.
- 2. Provide a regression table or `summary()` output for each model.
- 3. Using `beer_test`, generate **out-of-sample predictions**:
 - `predict log(price_floz)`
 - convert predictions back to `price_floz` using `exp()`
- 4. Evaluate predictive performance on `beer_test`:
 - report **RMSE** (required)

Requirements:

- Present results in a small table comparing Model 1–3 test RMSE.
- Briefly comment on which model predicts best and by how much.

Show answer

```
# Fit the models
m1 <- lm(data = beer_train,
         log(price_floz) ~ log(beer_floz) + brand + container + market)
m2 <- lm(data = beer_train,
         log(price_floz) ~ log(beer_floz) * brand + container + market)
m3 <- lm(data = beer_train,
         log(price_floz) ~ log(beer_floz) * brand * promo + container + market)

# Regression table
stargazer(m1, m2, m3,
          type = "html", # which requires the code chunk option: `#| results: asis`
          digits = 4)
```

Dependent variable:		
log(price_floz)		
(1)	(2)	(3)

log(beer_floz)	-0.1412*** (0.0012)	-0.1481*** (0.0021)	-0.1442*** (0.0023)
brandBUSCH_LIGHT	-0.2602*** (0.0028)	-0.2163*** (0.0219)	-0.1896*** (0.0231)
brandCOORS_LIGHT	-0.0028 (0.0024)	-0.0067 (0.0182)	0.0018 (0.0193)
brandMILLER_LITE	-0.0103*** (0.0022)	0.0710*** (0.0166)	0.1035*** (0.0178)
brandNATURAL_LIGHT	-0.3184*** (0.0025)	-0.6106*** (0.0174)	-0.5438*** (0.0186)
promo			-0.0786** (0.0368)
containerNON_REFILLABLE_BOTTLE	0.0522*** (0.0019)	0.0509*** (0.0019)	0.0522*** (0.0019)
marketALBANY	0.0302** (0.0125)	0.0322*** (0.0125)	0.0235* (0.0124)
marketATLANTA	0.0868*** (0.0101)	0.0855*** (0.0100)	0.0805*** (0.0099)
marketBALTIMORE	0.0983*** (0.0133)	0.1015*** (0.0132)	0.0901*** (0.0131)
marketBIRMINGHAM	0.1274*** (0.0103)	0.1340*** (0.0103)	0.1290*** (0.0102)
marketBOSTON	0.1307*** (0.0109)	0.1308*** (0.0108)	0.1265*** (0.0107)
marketCHARLOTTE	0.0284*** (0.0101)	0.0249** (0.0100)	0.0328*** (0.0100)
marketCHICAGO	-0.0044 (0.0095)	-0.0096 (0.0095)	-0.0047 (0.0094)
marketCINCINNATI	0.0849*** (0.0101)	0.0797*** (0.0100)	0.0768*** (0.0100)
marketCLEVELAND	0.0663*** (0.0102)	0.0618*** (0.0101)	0.0567*** (0.0100)
marketCOLUMBUS	0.0750*** (0.0095)	0.0720*** (0.0095)	0.0719*** (0.0094)
marketDALLAS	0.2110*** (0.0094)	0.2208*** (0.0094)	0.2241*** (0.0093)
marketDENVER	0.1328*** (0.0109)	0.1307*** (0.0109)	0.1413*** (0.0108)
marketDES_MOINES	0.1380*** (0.0112)	0.1353*** (0.0111)	0.1281*** (0.0110)
marketDETROIT	0.0904*** (0.0096)	0.0870*** (0.0096)	0.0877*** (0.0095)
marketEXURBAN_NJ	0.2309*** (0.0161)	0.2273*** (0.0161)	0.2175*** (0.0160)
marketEXURBAN_NY	0.1146*** (0.0220)	0.1118*** (0.0218)	0.1024*** (0.0217)
marketGRAND_RAPIDS	0.0906*** (0.0113)	0.0862*** (0.0112)	0.0833*** (0.0111)
marketHARTFORD-NEW_HAVEN	0.1483*** (0.0138)	0.1463*** (0.0137)	0.1432*** (0.0136)
marketHOUSTON	0.1217***	0.1189***	0.1216***

	(0.0097)	(0.0096)	(0.0095)
marketINDIANAPOLIS	0.0491***	0.0477***	0.0472***
	(0.0101)	(0.0101)	(0.0100)
marketJACKSONVILLE	0.1287***	0.1228***	0.1226***
	(0.0123)	(0.0123)	(0.0122)
marketKANSAS_CITY	0.0762***	0.0713***	0.0639***
	(0.0115)	(0.0115)	(0.0114)
marketLITTLE_ROCK	0.0976***	0.0935***	0.0883***
	(0.0127)	(0.0127)	(0.0126)
marketLOS_ANGELES	0.0389***	0.0328***	0.0407***
	(0.0097)	(0.0097)	(0.0096)
marketLOUISVILLE	0.0652***	0.0610***	0.0641***
	(0.0110)	(0.0109)	(0.0108)
marketMEMPHIS	0.1320***	0.1304***	0.1226***
	(0.0122)	(0.0121)	(0.0120)
marketMIAMI	0.1106***	0.1097***	0.1098***
	(0.0092)	(0.0091)	(0.0090)
marketMILWAUKEE	0.0277**	0.0262**	0.0277**
	(0.0113)	(0.0113)	(0.0112)
marketMINNEAPOLIS	0.1470***	0.1480***	0.1422***
	(0.0111)	(0.0110)	(0.0109)
marketNASHVILLE	0.1495***	0.1486***	0.1466***
	(0.0105)	(0.0105)	(0.0104)
marketNEW_ORLEANS-MOBILE	0.1362***	0.1269***	0.1198***
	(0.0108)	(0.0107)	(0.0106)
marketOKLAHOMA_CITY-TULSA	0.1501***	0.1467***	0.1385***
	(0.0110)	(0.0109)	(0.0109)
marketOMAHA	0.1321***	0.1292***	0.1292***
	(0.0104)	(0.0104)	(0.0103)
marketORLANDO	0.1047***	0.1032***	0.1043***
	(0.0103)	(0.0102)	(0.0101)
marketPHILADELPHIA	0.1089***	0.1080***	0.0935***
	(0.0128)	(0.0128)	(0.0127)
marketPHOENIX	0.1482***	0.1514***	0.1581***
	(0.0093)	(0.0093)	(0.0092)
marketPITTSBURGH	0.1027***	0.1008***	0.0932***
	(0.0136)	(0.0135)	(0.0134)
marketPORTLAND	0.1124***	0.1108***	0.1117***
	(0.0120)	(0.0119)	(0.0118)
marketRALEIGH-DURHAM	0.0874***	0.0881***	0.0814***
	(0.0103)	(0.0102)	(0.0101)
marketRICHMOND	0.0413***	0.0387***	0.0316***
	(0.0104)	(0.0103)	(0.0102)
marketRURAL_ALABAMA	0.1695***	0.1689***	0.1631***
	(0.0143)	(0.0142)	(0.0141)
marketRURAL_ARKANSAS	0.1722***	0.1742***	0.1646***
	(0.0185)	(0.0184)	(0.0183)
marketRURAL_CALIFORNIA	0.0539***	0.0501***	0.0516***
	(0.0109)	(0.0108)	(0.0107)
marketRURAL_COLORADO	0.2014***	0.2008***	0.2040***
	(0.0428)	(0.0425)	(0.0422)

marketRURAL_FLORIDA	0.0572*** (0.0121)	0.0478*** (0.0120)	0.0451*** (0.0119)
marketRURAL_GEORGIA	0.1375*** (0.0128)	0.1332*** (0.0127)	0.1268*** (0.0126)
marketRURAL_IDAHO	0.1546*** (0.0181)	0.1488*** (0.0180)	0.1486*** (0.0179)
marketRURAL_ILLINOIS	0.0179* (0.0102)	0.0166 (0.0102)	0.0130 (0.0101)
marketRURAL_INDIANA	0.0711*** (0.0124)	0.0729*** (0.0123)	0.0711*** (0.0122)
marketRURAL_IOWA	0.0617*** (0.0103)	0.0585*** (0.0103)	0.0538*** (0.0102)
marketRURAL_KANSAS	0.1308*** (0.0170)	0.1315*** (0.0169)	0.1214*** (0.0168)
marketRURAL_KENTUCKY	0.1529*** (0.0163)	0.1521*** (0.0163)	0.1485*** (0.0161)
marketRURAL_LOUISIANA	0.0827*** (0.0131)	0.0780*** (0.0131)	0.0665*** (0.0130)
marketRURAL_MAINE	0.0937*** (0.0134)	0.0914*** (0.0134)	0.0887*** (0.0133)
marketRURAL_MICHIGAN	0.0832*** (0.0111)	0.0801*** (0.0111)	0.0750*** (0.0110)
marketRURAL_MINNESOTA	0.1723*** (0.0195)	0.1767*** (0.0194)	0.1688*** (0.0192)
marketRURAL_MISSISSIPPI	0.0513*** (0.0140)	0.0459*** (0.0139)	0.0458*** (0.0138)
marketRURAL_MISSOURI	0.1059*** (0.0115)	0.1048*** (0.0115)	0.0970*** (0.0114)
marketRURAL_MONTANA	0.1199*** (0.0137)	0.1169*** (0.0137)	0.1239*** (0.0136)
marketRURAL_NEBRASKA	0.1476*** (0.0210)	0.1474*** (0.0208)	0.1418*** (0.0207)
marketRURAL_NEVADA	0.0451*** (0.0119)	0.0450*** (0.0118)	0.0425*** (0.0117)
marketRURAL_NEW_HAMPSHIRE	0.0654 (0.0441)	0.0489 (0.0439)	0.0435 (0.0435)
marketRURAL_NEW_MEXICO	0.1697*** (0.0129)	0.1657*** (0.0129)	0.1618*** (0.0128)
marketRURAL_NEW_YORK	-0.0264 (0.0566)	-0.0242 (0.0563)	-0.0398 (0.0558)
marketRURAL_NORTH_CAROLINA	0.0221** (0.0108)	0.0505*** (0.0108)	0.0372*** (0.0107)
marketRURAL_NORTH_DAKOTA	0.2335*** (0.0198)	0.2309*** (0.0197)	0.2307*** (0.0196)
marketRURAL_OHIO	0.1062*** (0.0158)	0.1030*** (0.0157)	0.0998*** (0.0156)
marketRURAL_OKLAHOMA	0.1318*** (0.0278)	0.1330*** (0.0276)	0.1210*** (0.0274)
marketRURAL_OREGON	-0.0057 (0.0346)	-0.0077 (0.0344)	-0.0097 (0.0341)
marketRURAL_PENNSYLVANIA	0.1265***	0.1262***	0.1147***

	(0.0143)	(0.0143)	(0.0141)
marketRURAL_SOUTH_CAROLINA	0.0582***	0.0568***	0.0582***
	(0.0100)	(0.0100)	(0.0099)
marketRURAL_SOUTH_DAKOTA	0.0842***	0.0823***	0.0763***
	(0.0178)	(0.0177)	(0.0176)
marketRURAL_TENNESSEE	0.1766***	0.1773***	0.1795***
	(0.0131)	(0.0130)	(0.0130)
marketRURAL_TEXAS	0.1722***	0.1710***	0.1667***
	(0.0096)	(0.0095)	(0.0094)
marketRURAL_VERMONT	0.0845***	0.0732***	0.0712***
	(0.0195)	(0.0195)	(0.0193)
marketRURAL_VIRGINIA	0.0237	0.0218	0.0164
	(0.0170)	(0.0170)	(0.0168)
marketRURAL_WASHINGTON	0.1003***	0.0990***	0.1112***
	(0.0139)	(0.0138)	(0.0138)
marketRURAL_WEST_VIRGINIA	-0.0399***	-0.0423***	-0.0527***
	(0.0149)	(0.0148)	(0.0147)
marketRURAL_WISCONSIN	0.0446***	0.0423***	0.0403***
	(0.0100)	(0.0100)	(0.0099)
marketRURAL_WYOMING	0.1915***	0.1900***	0.1833***
	(0.0317)	(0.0316)	(0.0313)
marketSACRAMENTO	0.0360***	0.0349***	0.0395***
	(0.0105)	(0.0105)	(0.0104)
marketSALT_LAKE_CITY	0.1050***	0.0986***	0.0954***
	(0.0142)	(0.0142)	(0.0141)
marketSAN_ANTONIO	0.1390***	0.1342***	0.1295***
	(0.0092)	(0.0091)	(0.0091)
marketSAN_DIEGO	0.0155	0.0146	0.0181
	(0.0117)	(0.0116)	(0.0115)
marketSAN_FRANCISCO	0.0713***	0.0685***	0.0741***
	(0.0107)	(0.0107)	(0.0106)
marketSEATTLE	0.1190***	0.1103***	0.1206***
	(0.0107)	(0.0106)	(0.0105)
marketST_LOUIS	0.0486***	0.0449***	0.0460***
	(0.0099)	(0.0099)	(0.0098)
marketSURBURBAN_NJ	-0.0162	-0.0186	-0.0315**
	(0.0131)	(0.0131)	(0.0130)
marketSURBURBAN_NY	0.1060***	0.1031***	0.1011***
	(0.0126)	(0.0126)	(0.0124)
marketSYRACUSE	-0.0340**	-0.0386***	-0.0491***
	(0.0146)	(0.0146)	(0.0145)
marketTAMPA	0.1103***	0.1071***	0.1069***
	(0.0090)	(0.0090)	(0.0089)
marketURBAN_NY	0.1615***	0.1616***	0.1597***
	(0.0113)	(0.0112)	(0.0111)
marketWASHINGTON_DC	0.1032***	0.0974***	0.0905***
	(0.0108)	(0.0107)	(0.0106)
log(beer_floz):brandBUSCH_LIGHT		-0.0077*	-0.0132***
		(0.0040)	(0.0042)
log(beer_floz):brandCOORS_LIGHT		0.0006	0.0001
		(0.0034)	(0.0036)

log(beer_floz):brandMILLER_LITE	-0.0151*** (0.0031)	-0.0211*** (0.0033)	
log(beer_floz):brandNATURAL_LIGHT	0.0542*** (0.0032)	0.0416*** (0.0035)	
log(beer_floz):promo		0.0061 (0.0067)	
brandBUSCH_LIGHT:promo		-0.1490** (0.0696)	
brandCOORS_LIGHT:promo		-0.1289** (0.0571)	
brandMILLER_LITE:promo		-0.2613*** (0.0513)	
brandNATURAL_LIGHT:promo		-0.3153*** (0.0507)	
log(beer_floz):brandBUSCH_LIGHT:promo		0.0280** (0.0123)	
log(beer_floz):brandCOORS_LIGHT:promo		0.0180* (0.0103)	
log(beer_floz):brandMILLER_LITE:promo		0.0472*** (0.0093)	
log(beer_floz):brandNATURAL_LIGHT:promo		0.0553*** (0.0092)	
Constant	-2.1776*** (0.0106)	-2.1388*** (0.0141)	-2.1491*** (0.0145)
Observations	48,314	48,314	48,314
R ²	0.5529	0.5575	0.5655
Adjusted R ²	0.5520	0.5566	0.5645
Residual Std. Error	0.1679 (df = 48214)	0.1670 (df = 48210)	0.1655 (df = 48200)
F Statistic	602.2622*** (df = 99; 48214)	589.7865*** (df = 103; 48210)	555.2074*** (df = 113; 48200)
Note:		p<0.1; p<0.05; p<0.01	

```
# stargazer(m1, m2, m3,
# type = "text") # to see a regression table just like in R Console
```

```
# Prediction and conversion
beer_test <- beer_test |>
  mutate(pred_1 = predict(m1, newdata = beer_test),
         pred_2 = predict(m2, newdata = beer_test),
         pred_3 = predict(m3, newdata = beer_test)
  ) |>
  mutate(price_floz_pred_1 = exp(pred_1),
         price_floz_pred_2 = exp(pred_2),
         price_floz_pred_3 = exp(pred_3)
  )
```

```
# RMSE
```

```
RMSEs <- beer_test |>
  mutate(
    resid_1 = log(price_floz) - pred_1,
    resid_2 = log(price_floz) - pred_2,
    resid_3 = log(price_floz) - pred_3
  ) |>
  mutate(
    resid_1_sq = resid_1^2,
    resid_2_sq = resid_2^2,
```

```

    resid_3_sq = resid_3^2
  ) |>
  summarize(
    rmse_1 = sqrt( mean(resid_1_sq) ),
    rmse_2 = sqrt( mean(resid_2_sq) ),
    rmse_3 = sqrt( mean(resid_3_sq) )
  )

```

```

RMSEs |>
  rmarkdown::paged_table()

```

rmse_1	rmse_2	rmse_3
<dbl>	<dbl>	<dbl>
0.170908	0.1701922	0.1690373

1 row

```

# RMSE with the same unit of price_floz

```

```

RMSEs_converted <- beer_test |>
  mutate(
    resid_1 = price_floz - exp(pred_1),
    resid_2 = price_floz - exp(pred_2),
    resid_3 = price_floz - exp(pred_3),
  ) |>
  mutate(
    resid_1_sq = resid_1^2,
    resid_2_sq = resid_2^2,
    resid_3_sq = resid_3^2
  ) |>
  summarize(
    rmse_1_converted = sqrt( mean(resid_1_sq) ),
    rmse_2_converted = sqrt( mean(resid_2_sq) ),
    rmse_3_converted = sqrt( mean(resid_3_sq) )
  )

```

```

RMSEs_converted |>
  rmarkdown::paged_table()

```

rmse_1_converted	rmse_2_converted	rmse_3_converted
<dbl>	<dbl>	<dbl>
0.008471677	0.008442086	0.008366237

1 row

Question 5 — Interpret market effects (Model 3)

Using **Model 3**, interpret the estimated coefficients for the following market indicators:

1. market_ALBANY
2. market_EXURBAN_NY
3. market_RURAL_NEW_YORK
4. market_SURBURBAN_NY
5. market_SYRACUSE
6. market_URBAN_NY

Requirements:

- Interpret each coefficient **relative to the reference market** (**BUFFALO-ROCHESTER**), holding other variables fixed.
- Since the outcome is in logs, convert each coefficient to an approximate percent difference using:
 - Approximation: $100 \times \beta$ (when $|\beta|$ is small)
 - Exact: $100 \times (\exp(\beta) - 1)$

Use whichever you prefer, but be consistent.

Show answer

```
m3_betas <- tidy(m3, conf.int = T) |>
  filter(
    term %in% c("marketALBANY", "marketEXURBAN_NY",
               "marketRURAL_NEW_YORK", "marketSUBURBAN_NY",
               "marketSYRACUSE", "marketURBAN_NY")
  ) |>
  mutate(
    stars = case_when(
      p.value < .01 ~ "***", # smallest p-values first
      p.value < .05 ~ "**",
      p.value < .1 ~ "*",
      TRUE ~ ""
    ),
    estimate_exp_1 = exp(estimate) - 1,
    .after = estimate
  ) |>
  mutate(estimate = round(estimate, 3),
         estimate_exp_1 = round(estimate_exp_1, 3)
  )

m3_betas |>
  relocate(stars, .after = estimate_exp_1) |>
  rmarkdown::paged_table()
```

term <chr>	estimate <dbl>	estimate_exp_1 <dbl>	stars <chr>	std.error <dbl>	statistic <dbl>	p.value <dbl>
marketALBANY	0.024	0.024	*	0.01237542	1.9012655	5.727321e-02
marketEXURBAN_NY	0.102	0.108	***	0.02165730	4.7269213	2.285925e-06
marketRURAL_NEW_YORK	-0.040	-0.039		0.05577772	-0.7144425	4.749570e-01
marketSUBURBAN_NY	0.101	0.106	***	0.01244790	8.1220719	4.690626e-16
marketSYRACUSE	-0.049	-0.048	***	0.01445557	-3.3990026	6.768681e-04
marketURBAN_NY	0.160	0.173	***	0.01111185	14.3731007	9.539258e-47

6 rows | 1-7 of 9 columns

- Below provides a list of beta estimates and exponential function of those beta estimates from Model 3:
 - market_ALBANY** ($\hat{\beta} = 0.024^*$, $e^{\hat{\beta}} = 1.024$)
Ceteris paribus, beer prices in Albany are 2.4% higher than in the Buffalo–Rochester market.
 - market_EXURBAN_NY** ($\hat{\beta} = 0.102^{***}$, $e^{\hat{\beta}} = 1.108$)
Ceteris paribus, beer prices in ExurbanNY are 10.8% higher than in the Buffalo–Rochester market.
 - market_RURAL_NEW_YORK** ($\hat{\beta} = -0.040$, $e^{\hat{\beta}} = 0.961$)
Ceteris paribus, beer prices in RuralNY are not statistically different than in the Buffalo–Rochester market.
 - market_SUBURBAN_NY** ($\hat{\beta} = 0.101^{***}$, $e^{\hat{\beta}} = 1.106$)
Ceteris paribus, beer prices in SuburbanNY are 10.6% higher than in the Buffalo–Rochester market.
 - market_SYRACUSE** ($\hat{\beta} = -0.049^{***}$, $e^{\hat{\beta}} = 0.952$)
Ceteris paribus, beer prices in Syracuse are 4.8% lower than in the Buffalo–Rochester market.

- `market_URBAN_NY` ($\hat{\beta} = 0.160^{***}$, $e^{\hat{\beta}} = 1.173$)

Ceteris paribus, beer prices in UrbanNY are 17.3% higher than in the Buffalo–Rochester market.

All CIs

```
m3_betas_ci95 <- tidy(m3, conf.int = T) |>
  mutate(ci = "95%")
m3_betas_ci90 <- tidy(m3, conf.int = T, conf.level = .9) |>
  mutate(ci = "90%")
m3_betas_ci99 <- tidy(m3, conf.int = T, conf.level = .99) |>
  mutate(ci = "99%")

m3_betas <- bind_rows(
  m3_betas_ci95,
  m3_betas_ci90,
  m3_betas_ci99
) |>
  filter(
    term %in% c("marketALBANY", "marketEXURBAN_NY",
               "marketRURAL_NEW_YORK", "marketSURBURBAN_NY",
               "marketSYRACUSE", "marketURBAN_NY")
  ) |>
  mutate(
    stars = case_when(
      p.value < .01 ~ "****", # smallest p-values first
      p.value < .05 ~ "***",
      p.value < .1 ~ "**",
      TRUE ~ ""
    ),
    estimate_exp_1 = exp(estimate)-1,
    .after = estimate
  ) |>
  mutate(estimate = round(estimate, 3),
         estimate_exp_1 = round(estimate_exp_1, 3)
  )

m3_betas |>
  relocate(stars, .after = estimate_exp_1) |>
  rmarkdown::paged_table()
```

term <chr>	estimate <dbl>	estimate_exp_1 <dbl>	stars <chr>	std.error <dbl>	statistic <dbl>	p.value <dbl>
marketALBANY	0.024	0.024	*	0.01237542	1.9012655	5.727321e-02
marketEXURBAN_NY	0.102	0.108	***	0.02165730	4.7269213	2.285925e-06
marketRURAL_NEW_YORK	-0.040	-0.039		0.05577772	-0.7144425	4.749570e-01
marketSURBURBAN_NY	0.101	0.106	***	0.01244790	8.1220719	4.690626e-16
marketSYRACUSE	-0.049	-0.048	***	0.01445557	-3.3990026	6.768681e-04
marketURBAN_NY	0.160	0.173	***	0.01111185	14.3731007	9.539258e-47
marketALBANY	0.024	0.024	*	0.01237542	1.9012655	5.727321e-02
marketEXURBAN_NY	0.102	0.108	***	0.02165730	4.7269213	2.285925e-06
marketRURAL_NEW_YORK	-0.040	-0.039		0.05577772	-0.7144425	4.749570e-01
marketSURBURBAN_NY	0.101	0.106	***	0.01244790	8.1220719	4.690626e-16

1-10 of 18 rows | 1-7 of 10 columns

Previous 1 2 Next

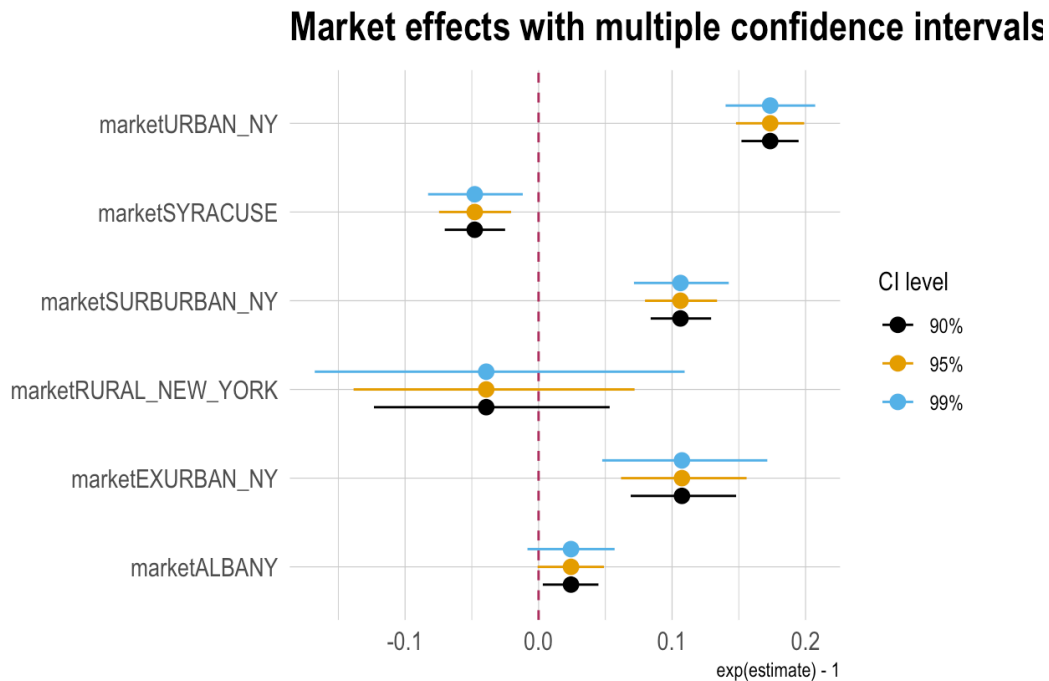
Coefficient plot

```
ggplot(
  m3_betas,
  aes(
    y = term,
    x = exp(estimate) - 1,
    xmin = exp(conf.low) - 1,
    xmax = exp(conf.high) - 1,
  )
)
```

```

    color = ci
  )
) +
  geom_vline(xintercept = 0, color = "maroon", linetype = 2) +
  geom_pointrange(
    position = position_dodge(width = 0.6)
  ) +
  labs(
    title = "Market effects with multiple confidence intervals",
    color = "CI level",
    y = ""
  )
)

```



Confidence Interval and True Beta

- The true value of β is 90% likely to be in the 90% confidence interval.
- The true value of β is 95% likely to be in the 95% confidence interval.
- The true value of β is 99% likely to be in the 99% confidence interval.

Question 6 — Volume sensitivity and promo (elasticities)

Across the three models, analyze **how sensitive price is to volume**, and how that differs by brand and promo:

1. In **Model 1**, explain what the coefficient on $\log(\text{beer_floz})$ means.
2. In **Model 2**, show how the volume sensitivity differs by brand.
3. In **Model 3**, explain how **promo** changes the sensitivity, and how the effect can differ by brand.

Requirements:

- Write the *expression* for the slope (marginal effect) of $\log(\text{beer_floz})$ for:
 - a particular brand when **promo** = FALSE

- the same brand when `promo = TRUE`
- Provide at least **one numeric example** using estimated coefficients from your Model 3 output (pick any one brand).

Show answer

- We should focus on the beta estimates for predictors with $\log(\text{beer_floz})$ to calculate the volume elasticity of beer price across brands.

Predictor	Model 1	Model 2	Model 3
<code>log_beer_floz</code>	-0.1412***	-0.1481***	-0.1442***
<code>brand_BUSCH_LIGHT_*_log_beer_floz</code>		-0.0077*	-0.0132***
<code>brand_COORS_LIGHT_*_log_beer_floz</code>		-0.0006	-0.0001
<code>brand_MILLER_LITE_*_log_beer_floz</code>		-0.0151***	-0.0211***
<code>brand_NATURAL_LIGHT_*_log_beer_floz</code>		0.0542***	0.0416***
<code>promo_True_*_log_beer_floz</code>			0.0061
<code>brand_BUSCH_LIGHT_promo_True_log_beer_floz</code>			0.0280**
<code>brand_COORS_LIGHT_promo_True_log_beer_floz</code>			0.0180*
<code>brand_MILLER_LITE_promo_True_log_beer_floz</code>			0.0472***
<code>brand_NATURAL_LIGHT_promo_True_log_beer_floz</code>			0.0553***

	Model 1	Model 2	Model 3 (no Promo)	Model 3 (with Promo)
BUD	-0.1412	-0.1481	-0.1442	-0.1442 = -0.1442 - 0
BUSCH	-0.1412	-0.1558 = -0.1481 - 0.0077	-0.1574 = -0.1442 - 0.0132	-0.1294 = -0.1442 - 0.0132 - 0 + 0.0280
COORS	-0.1412	-0.1481 = -0.1481 - 0	-0.1442 = -0.1442 - 0	-0.1262 = -0.1442 - 0 - 0 + 0.0180
MILLER	-0.1412	-0.1632 = -0.1481 - 0.0151	-0.1653 = -0.1442 - 0.0211	-0.1181 = -0.1442 - 0.0211 - 0 + 0.0472
NATURAL	-0.1412	-0.0939 = -0.1481 + 0.0542	-0.1026 = -0.1442 + 0.0416	-0.0473 = -0.1442 + 0.0416 - 0 + 0.0553

- We can extract a beta estimate using `coef()`. For example:

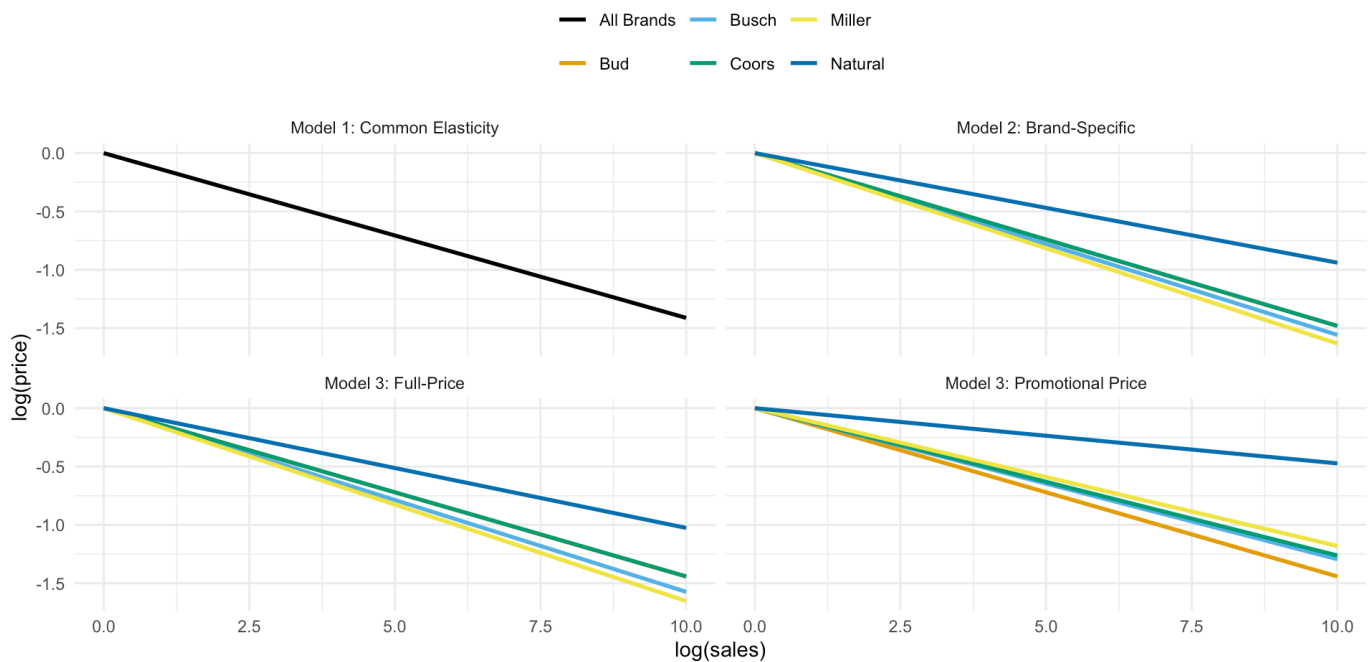
```
coef(m2)['log(beer_floz)'] + coef(m2)['log(beer_floz):COORS_LIGHT']
```

All else being equal:

- In Model 1
 - A 1% increase in sales volume (across any of the five brands) is associated with a 0.1412% decrease in price.
- In Model 2
 - A 1% increase in BUD sales volume is associated with a 0.1481% decrease in its price.
 - A 1% increase in BUSCH sales volume is associated with a 0.1558% decrease in its price.
 - A 1% increase in COORS sales volume is associated with a 0.1481% decrease in its price.
 - A 1% increase in MILLER sales volume is associated with a 0.1632% decrease in its price.

- A 1% increase in NATURAL sales volume is associated with a 0.0939% decrease in its price.
- In Model 3 (no Promo)
 - A 1% increase in BUD sales volume is associated with a 0.1442% decrease in its price.
 - A 1% increase in BUSCH sales volume is associated with a 0.1574% decrease in its price.
 - A 1% increase in COORS sales volume is associated with a 0.1442% decrease in its price.
 - A 1% increase in MILLER sales volume is associated with a 0.1653% decrease in its price.
 - A 1% increase in NATURAL sales volume is associated with a 0.1026% decrease in its price.
- In Model 3 (with Promo)
 - A 1% increase in BUD sales volume is associated with a 0.1442% decrease in its price.
 - A 1% increase in BUSCH sales volume is associated with a 0.1294% decrease in its price.
 - A 1% increase in COORS sales volume is associated with a 0.1262% decrease in its price.
 - A 1% increase in MILLER sales volume is associated with a 0.1181% decrease in its price.
 - A 1% increase in NATURAL sales volume is associated with a 0.0473% decrease in its price.

► Code



Across all three models, larger package sizes are consistently associated with lower price per fl oz, confirming a standard quantity-discount pattern. The estimated elasticities are all small and negative, ranging roughly from -0.05% to -0.17% per 1% increase in volume.

- In Model 1, a single elasticity of -0.1412% applies to all brands.
- Model 2 reveals meaningful brand-level differences: NATURAL has the smallest discount (-0.0939%), while MILLER has the largest (-0.1632%), with BUD and COORS tied in the middle (-0.1481%).
- Model 3 (no promo) tells a broadly similar story, though the estimates shift slightly.
- The more striking finding appears in Model 3 under promotion: for most brands, the quantity discount becomes noticeably smaller when an item is on promotion. NATURAL drops from -0.1026% to -0.0473% , MILLER from -0.1653% to -0.1181% , and COORS from -0.1442% to -0.1262% .
 - BUD is the exception, with its elasticity unchanged at -0.1442% regardless of promotional status.
 - This suggests that promotions compress the size-price relationship for most brands, but BUD's per-unit pricing scales with volume in the same way whether or not it is on promotion.

Question 7 — Residual diagnostics

For each model (1–3), create a residual plot:

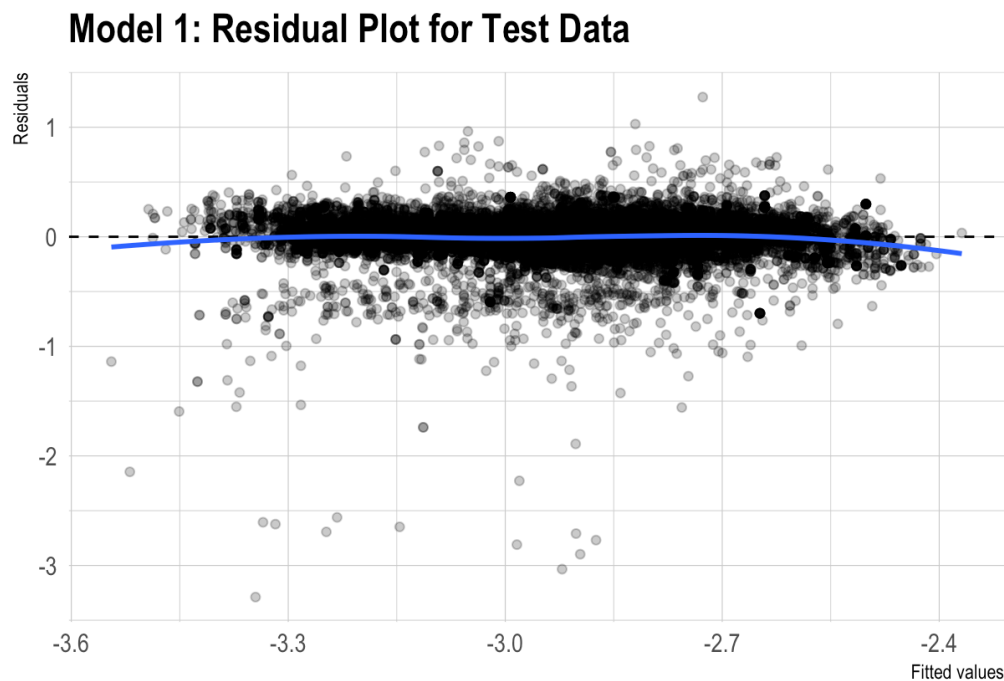
- x-axis: fitted values
- y-axis: residuals
- include a horizontal line at 0

Then answer:

- Are residuals centered around 0 on average?
- Do you see systematic patterns (curvature, fan shape / heteroskedasticity, clusters, outliers)?
- Which model appears to have the best residual behavior?

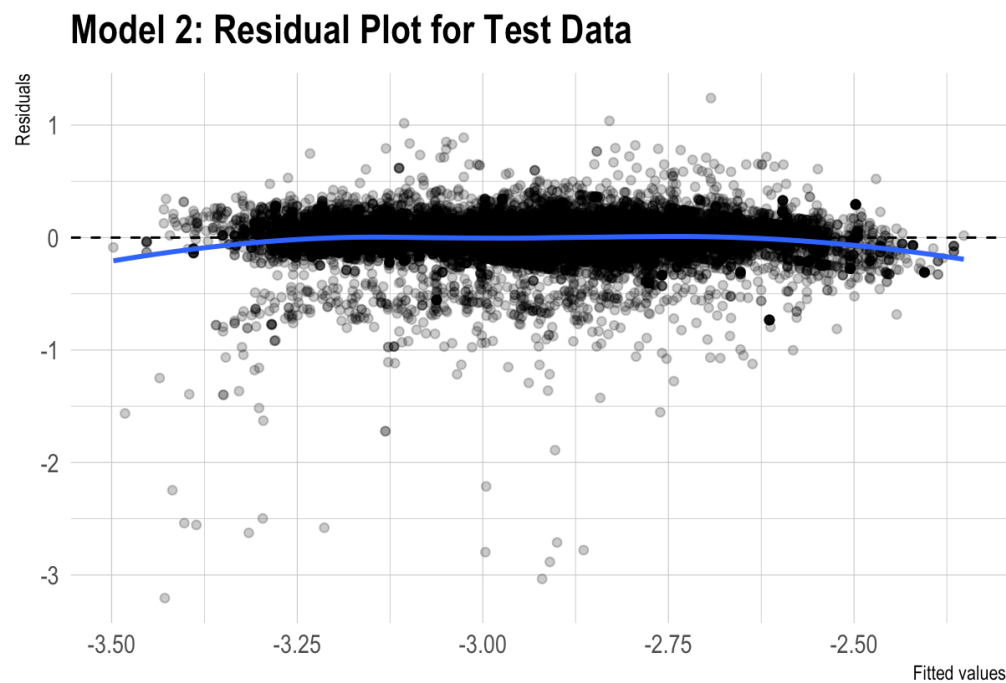
Show answer

```
m1 |>
  augment(newdata = beer_test) |>
  ggplot(aes(x = .fitted,
             y = .resid)) +
  geom_point(alpha = 0.25) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  geom_smooth(se = FALSE, method = "loess") +
  labs(
    x = "Fitted values",
    y = "Residuals",
    title = "Model 1: Residual Plot for Test Data"
  )
```



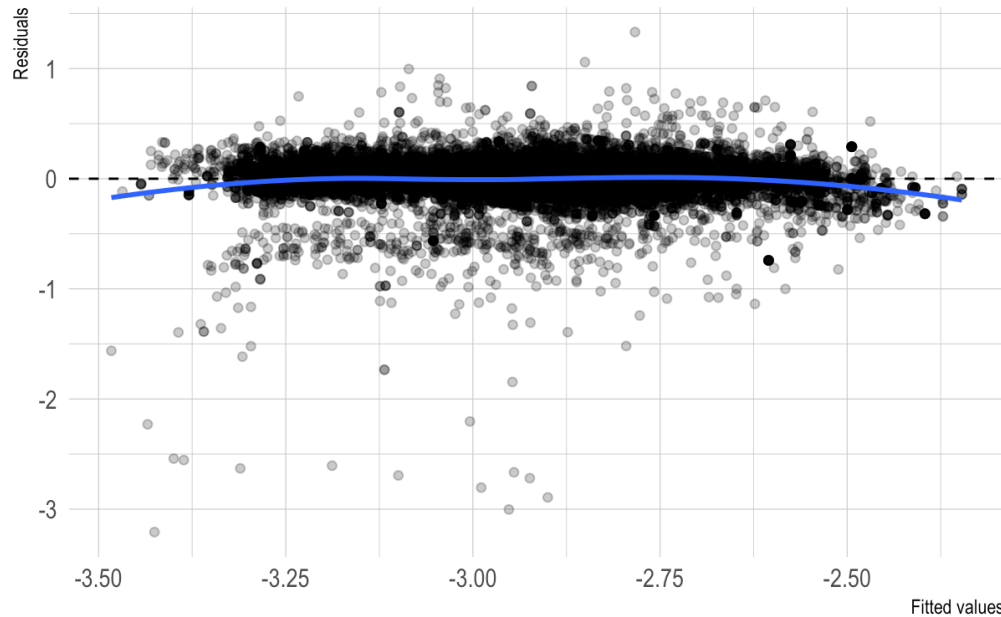
```
m2 |>
  augment(newdata = beer_test) |>
  ggplot(aes(x = .fitted,
             y = .resid)) +
  geom_point(alpha = 0.25) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  geom_smooth(se = FALSE, method = "loess") +
```

```
labs(
  x = "Fitted values",
  y = "Residuals",
  title = "Model 2: Residual Plot for Test Data"
)
```



```
m3 |>
augment(newdata = beer_test) |>
ggplot(aes(x = .fitted,
  y = .resid)) +
geom_point(alpha = 0.25) +
geom_hline(yintercept = 0, linetype = "dashed") +
geom_smooth(se = FALSE, method = "loess") +
labs(
  x = "Fitted values",
  y = "Residuals",
  title = "Model 3: Residual Plot for Test Data"
)
```

Model 3: Residual Plot for Test Data



- Are the residuals centered around 0 (on average)?
 - Yes. The residuals appear to be roughly balanced above and below 0, with no clear shift away from zero.
- Do you see systematic patterns (curvature, fan shape / heteroskedasticity, clusters, outliers)?
 - No. I don't see obvious curvature or a fan-shaped spread, and there are no clear clusters or extreme outliers that stand out.
 - Model 1 has a wider range of fitted values than Models 2 and 3, but this is minor.
- Which model appears to have the best residual behavior?
 - The residual plots look very similar across models.
 - None clearly dominates based on residual behavior alone, so they appear **comparably good** on this diagnostic.

Question 8 — Choose a preferred model

Which model do you prefer most and why? Your answer must reference at least **two** of the following:

- out-of-sample performance (test RMSE)
- interpretability / simplicity
- whether interactions are substantively meaningful
- residual plot patterns

Show answer

I prefer Model 3 for three simple reasons:

1. Realistic Market Setting
 - It captures both the unique sensitivity of each brand and how that sensitivity changes when a beer is on promotion.
2. Practical Pricing Strategies
 - By distinguishing full-price from promotional periods, it tells you exactly how much to adjust each brand's price under each scenario.
3. Better Fit with the lowest MSE on test data

Part 2. Quarto Blogging



Use the following data frame for Quarto Blogging:

```
#| warning: false

library(tidyverse)
ice_cream <- read_csv("https://bcdanl.github.io/data/ben-and-jerry-cleaned.csv")
```

priceper1	flavor_descr	size1_descr	household_id	household_income
<dbl>	<chr>	<chr>	<dbl>	<dbl>
3.410000	CAKE BATTER	16.0 MLOZ	2001456	130000
3.500000	VAN CARAMEL FUDGE	16.0 MLOZ	2001456	130000
3.500000	VAN CARAMEL FUDGE	16.0 MLOZ	2001456	130000
3.000000	W-N-C-P-C	16.0 MLOZ	2001637	70000
3.990000	AMERICONE DREAM	16.0 MLOZ	2002791	130000
3.890000	ONE CSK BROWNIE	16.0 MLOZ	2002791	130000
3.890000	PUMPKIN CSK	16.0 MLOZ	2002791	130000
2.140000	CHC ALMOND NOUGAT	16.0 MLOZ	2001878	110000
3.500000	PISTACHIO PISTACHIO	16.0 MLOZ	2002721	210000
3.000000	PISTACHIO PISTACHIO	16.0 MLOZ	2002721	210000
4.000000	PISTACHIO PISTACHIO	16.0 MLOZ	2002721	210000
2.990000	CHERRY GRCA	16.0 MLOZ	2002800	70000
2.500000	CHERRY GRCA	16.0 MLOZ	2002800	70000
2.270000	NEW YORK SUPER FUDGE CHUNK	16.0 MLOZ	2002800	70000
2.550000	BANANA SPLIT	16.0 MLOZ	2000358	70000